

ICT1242 Web Development Practicum

Group 05 – Advanced Features Meeting Report

Hospital Appointment, Admission, and Billing System

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Group	Group 05 (TG2080, TG2086, TG2088, TG2095, TG2096, TG2097, TG2112, TG2127, TG2136)
Meeting type	Internal Planning – Advanced System Feature Discussion
Subject	Advanced module features beyond base proposal

1. Purpose of This Report

This report documents the advanced system features discussed at today's Group 05 planning meeting. During our first meeting, we defined the core modules for the Hospital Appointment, Admission, and Billing System. This session focused on identifying enhanced features that will elevate the system beyond a basic implementation. The features discussed are grouped into the following advanced modules:

- Advanced Appointment Flow with OPD Sync
- Emergency Ambulance Dispatch System
- Advanced Ward and Room Management
- Theatre (Operation) Management
- Advanced Pharmacy and Drug Management
- Enhanced Billing – Full and Advance Payment
- Admin Dashboard Enhancements

2. Advanced Appointment Module

2.1 Patient Self-Registration and Appointment Booking

Patients can register and log in through the web portal. Once logged in they can:

- Select a doctor from the system and view available time slots
- Book an appointment – upon confirmation the system assigns a unique appointment number, date, and time
- Reschedule or cancel existing appointments from their patient dashboard
- View full appointment history

2.2 OPD Walk-in to Online Appointment Synchronisation

A key advanced requirement is that appointments created at the hospital OPD counter must sync with the online system in real time. This means:

- A receptionist creating a walk-in appointment at the OPD counter will see the same doctor's calendar as online patients

- Double-booking is prevented across both channels
- Patients registered online can also receive OPD-generated appointments in their portal
- All appointment records share a single database table regardless of creation source

Feature	Description
Online booking	Patient selects doctor, date, and time via web portal
OPD sync	OPD counter and online system share one live calendar
Appointment no.	Unique appointment number auto-generated on confirmation
Reschedule/cancel	Patient can reschedule or cancel from dashboard
Doctor filter	Filter doctors by specialization before booking
Add / Edit / Delete	Admin and reception can manage all appointment records

3. Emergency Ambulance Dispatch System

3.1 Overview

The emergency module allows patients to request an ambulance directly from the hospital website. This is a two-step process: the patient shares their location and contact details, and the hospital dispatcher calls back to confirm and dispatch.

3.2 Patient Side – Emergency Request

- Patient opens the Emergency page on the hospital website
- System captures the patient's current GPS location (browser geolocation API)
- Patient enters a short description of the emergency and their phone number
- Request is submitted – a unique emergency ticket number is generated
- Patient sees a confirmation message: 'Your request has been received. The hospital will call you shortly.'

3.3 Hospital Dispatcher Side

- Dispatcher sees all incoming emergency requests in real time on the admin panel
- Each request shows: patient name/description, phone number, GPS location (map link), and timestamp
- Dispatcher calls the patient's phone number to confirm situation
- Dispatcher selects the nearest available ambulance and assigns it to the request
- System records the assigned ambulance, driver, and estimated departure time
- Once ambulance departs, status changes to 'En Route'

3.4 Emergency Ward Notification

Once an emergency is confirmed and the ambulance is dispatched, the Emergency Ward is automatically notified in the system so staff can prepare for the incoming patient.

Feature	Description
Patient request	Web form with GPS location capture and phone number
Dispatcher view	Real-time list of all incoming emergency requests

Phone callback	Dispatcher calls patient to confirm and assess severity
Ambulance assign	Select and assign available ambulance to the request
Status tracking	Request status: Pending → Dispatched → En Route → Arrived
Ward alert	Emergency ward notified automatically on dispatch

4. Advanced Ward and Room Management

4.1 Room Categories

Unlike a standard ward system, this hospital uses private room-based accommodation rather than open wards. The following room types are available:

- Semi-Private Room – Shared room with limited beds (2–4 beds)
- Private Room – Single occupancy, full privacy
- Children's Room – Paediatric ward, child-appropriate facilities
- ICU (Intensive Care Unit) – High-dependency monitoring

4.2 Room Allocation and Patient Admissions

- On admission, receptionist assigns the patient to an appropriate room category
- System shows real-time availability of each room type
- Room allocation records the admission date, assigned doctor, and room number
- Patient's dietary needs are recorded at admission

4.3 Food and Meal Management

The system handles two types of patient meals:

- Hospital-provided meals: Patients on the standard hospital diet receive meals from the hospital kitchen. Meal plans are generated from the dietary requirements entered at admission.
- Doctor-prescribed diet: If the patient requires a special diet ordered by the doctor, that prescription is stored in the system and the kitchen team is notified.

4.4 Labour and Maternity – Theatre-to-Ward Link

For maternity cases, when a patient is brought to the Labour Theatre for delivery, the system creates a link between the patient's record and the Labour Theatre. After delivery:

- The baby's data is captured (date of birth, weight, birth details)
- A new patient profile is created for the newborn, linked to the mother
- The newborn is assigned to the Children's room
- A small indicator light status is recorded for the newborn (e.g. stable / requires monitoring)

5. Theatre (Operation) Management

5.1 Operation Record

When a patient requires an operation, the system creates an Operation Record containing:

- Operation type (e.g. appendectomy, caesarean, orthopaedic, etc.)

- Lead surgeon (doctor assigned to perform the operation)
- Assisting doctors / anaesthesiologist
- Patient details and pre-op notes
- Scheduled theatre date and time
- Post-op notes and recovery instructions

5.2 Theatre Scheduling

- Admin can view the Theatre calendar and schedule operations
- System prevents double-booking of the theatre at the same time
- After operation, patient is automatically transferred to the relevant ward (e.g. ICU post-op)

5.3 Theatre-to-Billing Link

All theatre charges (surgical fees, theatre use fee, anaesthesia) are automatically added to the patient's billing record once the operation is confirmed.

Feature	Description
Operation record	Type, surgeon, assistants, date/time, notes
Theatre calendar	Schedule and view upcoming operations
Auto ward transfer	Patient moved to appropriate ward after operation
Auto billing	Theatre and surgical fees added to patient invoice
Labour link	Maternity operations linked to newborn profile creation

6. Advanced Pharmacy and Drug Management

6.1 Drug Inventory

- Pharmacists can add, edit, and delete drugs in the system
- Each drug entry includes: name, category, stock quantity, unit price, and reorder level
- System alerts when stock falls below the reorder level

6.2 Prescription Dispensing

- When a doctor prescribes medication, a prescription is created in the system linked to the patient's record
- Pharmacist views pending prescriptions and dispenses accordingly
- Stock quantity is automatically deducted when a prescription is dispensed
- A dispensing record is created for audit purposes

6.3 Billing Integration

All dispensed medications are automatically added as line items to the patient's billing invoice. No manual entry is required.

Feature	Description
Drug catalogue	Add, edit, delete drugs with stock and pricing

Stock alerts	Low stock warning when quantity hits reorder level
Prescription link	Doctor prescriptions appear in pharmacist queue
Auto dispensing	Stock deducted on dispensing, record kept
Billing auto-add	Dispensed drugs automatically billed to patient
Dispensing history	Full audit trail of all dispensed medications

7. Enhanced Billing and Payment

7.1 Payment Options

The billing module supports two payment modes:

- Full Payment – Patient settles the complete outstanding balance at checkout
- Advance Payment – Patient makes a partial payment at any point during admission; the remaining balance is tracked and due at discharge

7.2 Invoice Components

A patient invoice is automatically built from all linked modules:

- Consultation / appointment fees
- Room / ward charges (per day)
- Theatre and surgical fees
- Pharmacy / medication charges
- Laboratory or investigation fees (if applicable)
- Any other miscellaneous charges added by reception

7.3 Payment Records

- Each payment is recorded with date, amount, payment method, and received-by staff member
- Balance updates in real time after each advance payment
- A printable receipt is generated for every payment
- Final receipt issued on full settlement at discharge

8. Admin Dashboard Enhancements

The administrator has a central dashboard that provides full visibility over all system activity. Key dashboard features include:

Feature	Description
Doctor management	Add, edit, and deactivate doctor profiles and specializations
System overview	Live counts: today's appointments, admitted patients, pending bills
Revenue summary	Daily/weekly/monthly revenue from billing records
Emergency monitor	Real-time view of all incoming emergency requests
Ward occupancy	Room-by-room availability across all ward types
Appointment report	Appointments by doctor, by date, by status

Payment tracking	Settled vs outstanding invoices, advance payments pending
Staff management	Add and manage reception and pharmacy staff accounts
Audit log	Track all add/edit/delete actions performed in the system

9. Summary of Advanced Modules

Module	Priority	Key Highlight
Advanced Appointment + OPD Sync	High	Unified calendar for online and walk-in patients
Emergency Ambulance Dispatch	High	GPS-based request + dispatcher callback + ward alert
Advanced Ward & Room Types	High	4 room categories, dietary management, maternity link
Theatre / Operation Management	Medium	Operation scheduling, auto billing, auto ward transfer
Pharmacy & Drug Management	Medium	Drug catalogue, prescription dispensing, stock alerts
Enhanced Billing (Full/Advance)	High	Advance payment tracking, auto invoice, receipts
Admin Dashboard Enhancements	High	Full system overview, revenue, audit log

10. Agreed Next Steps

The following actions were agreed at the end of today's meeting:

1. Update the project proposal to include the advanced modules listed in this report.
2. Revise the database schema to include new tables: emergency_requests, ambulances, theatre_operations, pharmacy_drugs, prescriptions, newborn_records.
3. Assign development responsibilities for each advanced module to group members.
4. Begin frontend wireframes for the Emergency page, Ward Management page, and Theatre Scheduling page.
5. Next group meeting to be scheduled within 7 days to review wireframes and finalise task allocation.